EE1007
Introduction to Computer Science Laboratory

Lab 2
Variables and Arithmetic

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Lab 2.1
#P110003: Temperature unit conversion (30%)

- In this program, you are asked to convert the temperature value from Celsius degree to Fahrenheit degree
- Relationship between Celsius and Fahrenheit:
  - F = 9/5 * C + 32
- The program will take a temperature of Celsius degree (a real number) as an input
- The program will print the corresponding value in Fahrenheit degree on the screen

Lab 2.1
#P110003: Temperature unit conversion (30%)

- The screen dialog should appear as follows:

```
Please input temperature value in Celsius degree: 25.5
The corresponding value in Fahrenheit degree is: 77.9
Process returned 0 (0x0)   execution time : 2.624 s
Press any key to continue.
```


Lab 2.2
#P110004: Arithmetic (30%)

- In this program, you need to perform arithmetic operations on two numbers
- Your program will take 2 different integers from keyboard
- Then your program will perform the corresponding operations based on the selection input:
  - 1 for summation
  - 2 for multiplication
  - 3 for finding the larger number
  - 4 for finding the smaller number
- Finally, your program will output the corresponding answer

Lab 2.2
#P110004: Arithmetic (30%)

◆The screen dialog should appear as follows:

```
Please input 2 different integers:
8
7
Please input the operation:
3
Larger number is 8

Process returned 0 (0x0)   execution time : 4.414 s
Press any key to continue.
```


Lab 2.2
#P110004: Arithmetic (30%)

Sample Input 1

```
Please input 2 different integers:
8
7
Please input the operation:
1
```

Sample Output 1

```
Summation is 15
```

Sample Input 2

```
Please input 2 different integers:
8
7
Please input the operation:
2
```

Sample Output 2

```
Multiplication is 56
```

Sample Input 3

```
Please input 2 different integers:
8
7
Please input the operation:
3
```

Sample Output 3

```
Larger number is 8
```

Sample Input 4

```
Please input 2 different integers:
8
7
Please input the operation:
4
```

Sample Output 4

```
Smaller number is 7
```

Lab 2.3
#P110005: BMI Calculator (40%)

- This program will calculate BIM for given weight and height
- The BMI is calculated as follows
  - BMI= $\frac{\text{weight(kg)}}{\text{height}^2(\text{m}^2)}$
- Your program will take two inputs, weight and height, from keyboard
- Then your program will output the corresponding BMI on the screen
- If your BMI is smaller than 18.5, “You need to eat more” will be printed on screen
- If your BMI is larger than (include) 18.5 and smaller than 24, “You are healthy” will be printed on screen
- If your BMI is larger than (include) 24, “You need to control your diet” will be printed on screen

Lab 2.3
#P110005: BMI Calculator (40%)

- The screen dialog should appear as follows:

```
Please enter your weight: 74.1
Please enter your height: 1.706
Your BMI is 25.4601
You need to control your diet
Process returned 0 (0x0)   execution time : 3.862 s
Press any key to continue.
```

Evaluation

- Lab2.1(30%)
  - 5 testcases, 6% each
- Lab2.2(30%)
  - 5 testcases, 6% each
- Lab2.3(40%)
  - 5 testcases, 8% each

How to Submit and Demo?

- Please do all the three problems and submit your codes to On-line Judge (OJ) system for evaluations
- You can try as many times as possible until your code is accepted by OJ
- When you finished the three problems or you decide to leave, BE SURE to find a TA for demo
  - TA will check your status on OJ and you code, log your attendance and score you obtain
- Upload all your source codes to ee-class and you are good to go!
  - BE SURE to follow the naming rule!!!
  - 學號_姓名_LabX.X.cpp (ex: 9862534_陳聿廣_Lab2.1.cpp)

Find a TA for demo	Upload your file to ee-class	Penalty
Before you leave class room	Before 21:00 Upload correct files and correctly follow naming rules	You are good!
Before you leave class room	Before 21:00 Upload wrong files or Not follow naming rules	Your submission will be returned. 20% penalty if you correct it within 72 hrs. Otherwise 0 point.
Before you leave class room	None	20% penalty if you upload it within 72 hrs. Otherwise 0 point.
None	Before 21:00 Upload correct file and correctly follow naming rules	20% penalty If you find a TA for make-up demo at next class. Otherwise 0 point.
None	Before 21:00 Upload wrong files or Not follow naming rules	Your submission will be returned. 20% penalty if you “correct it within 72 hrs” and “find a TA for make-up demo at next class.” Otherwise 0 point.
None	None	20% penalty if you “upload it within 72 hrs” and “find a TA for make-up demo at next class.” Otherwise 0 point.



離開教室前…

找助教

上傳Code

關電腦



Problem Discussion

- If you are able to type Chinese in you Code::Blocks…
  - 方法一

點選工具列
「Setting」→「Editor」→「General settings」→
「Encoding settings」→「WINDOWS-950(此為預設)」→
選擇「As default encoding」

  - 方法二

Setting -> Compiler settings -> Other settings -> Compiler output to be ignored:
加入
"-finput-charset=UTF-8"
"-fexec-charset=WINDOWS-950" (P.S. WINDOWS-90=Big5)
這兩行即可。





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